

Data Mining and Machine Learning

A Master Course

Master program on Data Science

Tran Trong Hieu¹

¹VNU - University of Engineering and Technology

Chapter 6: High-dimensional Data

High-dimensional Space

Let $\mathbf{D}$ be a $n \times d$ data matrix. In data mining typically the data is very high dimensional. Understanding the nature of high-dimensional space, or *hyperspace*, is very important, especially because it does not behave like the more familiar geometry in two or three dimensions.

Hyper-rectangle: The data space is a d -dimensional *hyper-rectangle*

$$R_d = \prod_{j=1}^d [\min(X_j), \max(X_j)]$$

where $\min(X_j)$ and $\max(X_j)$ specify the range of X_j .

Hypercube: Assume the data is centered, and let m denote the maximum attribute value

$$m = \max_{j=1}^d \max_{i=1}^n \{ |x_{ij}| \}$$

The data hyperspace can be represented as a *hypercube*, centered at 0, with all sides of length $l = 2m$, given as

$$H_d(l) = \left\{ \mathbf{x} = (x_1, x_2, \dots, x_d)^T \mid \forall i, x_i \in [-l/2, l/2] \right\}$$

The *unit hypercube* has all sides of length $l = 1$, and is denoted as $H_d(1)$.

Hypersphere

Assume that the data has been centered, so that $\mu = 0$. Let r denote the largest magnitude among all points:

$$r = \max_i \{ \|\mathbf{x}_i\| \}$$

The data hyperspace can be represented as a d -dimensional *hyperball* centered at 0 with radius r , defined as

$$B_d(r) = \{ \mathbf{x} \mid \|\mathbf{x}\| \leq r \} \text{ or } B_d(r) = \left\{ \mathbf{x} = (x_1, x_2, \dots, x_d) \mid \sum_{j=1}^d x_j^2 \leq r^2 \right\}$$

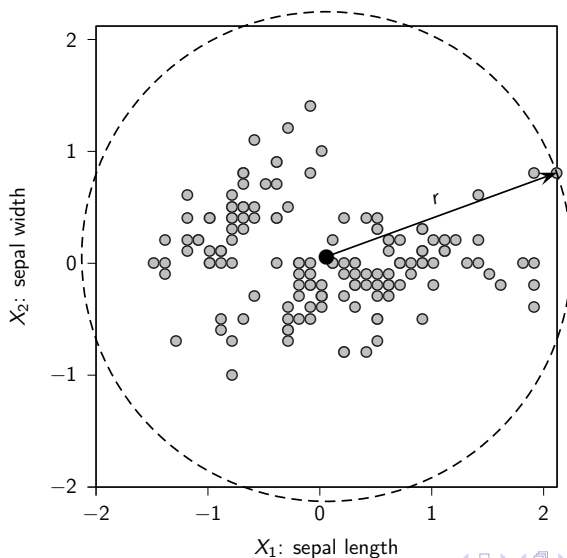
The surface of the hyperball is called a *hypersphere*, and it consists of all the points exactly at distance r from the center of the hyperball

$$S_d(r) = \{ \mathbf{x} \mid \|\mathbf{x}\| = r \}$$

or $S_d(r) = \left\{ \mathbf{x} = (x_1, x_2, \dots, x_d) \mid \sum_{j=1}^d (x_j)^2 = r^2 \right\}$

Iris Data Hyperspace: Hypercube and Hypersphere

$l = 4.12$ and $r = 2.19$



High-dimensional Volumes

Hypercube: The volume of a hypercube with edge length l is given as

$$\text{vol}(H_d(l)) = l^d$$

Hypersphere The volume of a hyperball and its corresponding hypersphere is identical
The volume of a hypersphere is given as

$$\text{In 1D: } \text{vol}(S_1(r)) = 2r \quad \text{In 2D: } \text{vol}(S_2(r)) = \pi r^2 \quad \text{In 3D: } \text{vol}(S_3(r)) = \frac{4}{3}\pi r^3$$

In d -dimensions:

$$\text{vol}(S_d(r)) = K_d r^d = \left(\frac{\pi^{\frac{d}{2}}}{\Gamma(\frac{d}{2} + 1)} \right) r^d$$

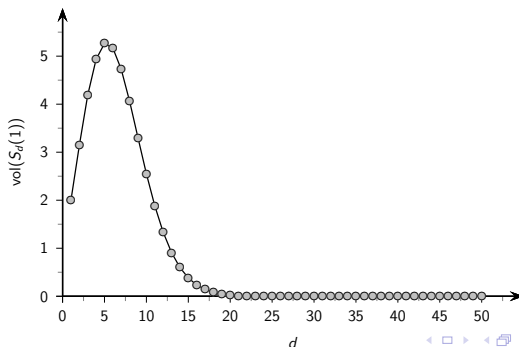
where

$$\Gamma\left(\frac{d}{2} + 1\right) = \begin{cases} \left(\frac{d}{2}\right)! & \text{if } d \text{ is even} \\ \sqrt{\pi} \left(\frac{d!!}{2^{(d+1)/2}}\right) & \text{if } d \text{ is odd} \end{cases}$$

Volume of Unit Hypersphere

With increasing dimensionality the hypersphere volume first increases up to a point, and then starts to decrease, and ultimately vanishes. In particular, for the unit hypersphere with $r = 1$,

$$\lim_{d \rightarrow \infty} \text{vol}(S_d(1)) = \lim_{d \rightarrow \infty} \frac{\pi^{\frac{d}{2}}}{\Gamma(\frac{d}{2} + 1)} \rightarrow 0$$



Hypersphere Inscribed within Hypercube

Consider the space enclosed within the largest hypersphere that can be accommodated within a hypercube (which represents the dataspace).

The ratio of the volume of the hypersphere of radius r to the hypercube with side length $l = 2r$ is given as

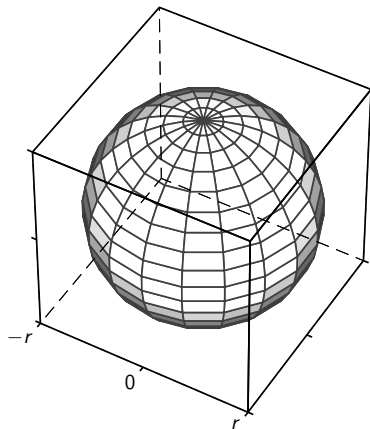
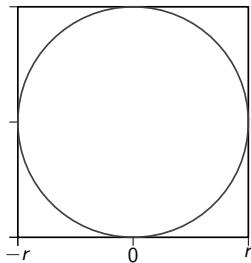
$$\text{In 2 dimensions: } \frac{\text{vol}(S_2(r))}{\text{vol}(H_2(2r))} = \frac{\pi r^2}{4r^2} = \frac{\pi}{4} = 78.5\%$$

$$\text{In 3 dimensions: } \frac{\text{vol}(S_3(r))}{\text{vol}(H_3(2r))} = \frac{\frac{4}{3}\pi r^3}{8r^3} = \frac{\pi}{6} = 52.4\%$$

$$\text{In } d \text{ dimensions: } \lim_{d \rightarrow \infty} \frac{\text{vol}(S_d(r))}{\text{vol}(H_d(2r))} = \lim_{d \rightarrow \infty} \frac{\pi^{d/2}}{2^d \Gamma(\frac{d}{2} + 1)} \rightarrow 0$$

As the dimensionality increases, most of the volume of the hypercube is in the “corners,” whereas the center is essentially empty.

Hypersphere Inscribed inside a Hypercube

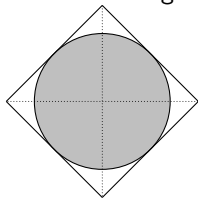


Conceptual View of High-dimensional Space

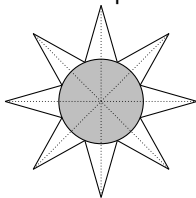
Two, three, four, and higher dimensions

All the volume of the hyperspace is in the corners, with the center being essentially empty.

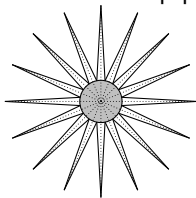
High-dimensional space looks like a rolled-up porcupine!



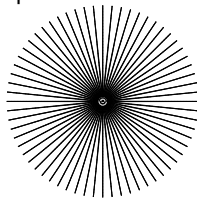
(a) 2D



(b) 3D



(c) 4D



(d) dD

Volume of a Thin Shell

The volume of a thin hypershell of width ϵ is given as

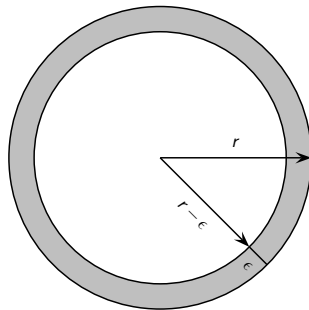
$$\begin{aligned}\text{vol}(S_d(r, \epsilon)) &= \text{vol}(S_d(r)) - \text{vol}(S_d(r - \epsilon)) \\ &= K_d r^d - K_d (r - \epsilon)^d.\end{aligned}$$

The ratio of volume of the thin shell to the volume of the outer sphere:

$$\frac{\text{vol}(S_d(r, \epsilon))}{\text{vol}(S_d(r))} = \frac{K_d r^d - K_d (r - \epsilon)^d}{K_d r^d} = 1 - \left(1 - \frac{\epsilon}{r}\right)^d$$

As d increases, we have

$$\lim_{d \rightarrow \infty} \frac{\text{vol}(S_d(r, \epsilon))}{\text{vol}(S_d(r))} = \lim_{d \rightarrow \infty} 1 - \left(1 - \frac{\epsilon}{r}\right)^d \rightarrow 1$$



Diagonals in Hyperspace

Consider a d -dimensional hypercube, with origin $0_d = (0_1, 0_2, \dots, 0_d)$, and bounded in each dimension in the range $[-1, 1]$. Each “corner” of the hyperspace is a d -dimensional vector of the form $(\pm 1_1, \pm 1_2, \dots, \pm 1_d)^T$.

Let $\mathbf{e}_i = (0_1, \dots, 1_i, \dots, 0_d)^T$ denote the d -dimensional canonical unit vector in dimension i , and let $\mathbf{1}$ denote the d -dimensional diagonal vector $(1_1, 1_2, \dots, 1_d)^T$.

Consider the angle θ_d between the diagonal vector $\mathbf{1}$ and the first axis $\mathbf{e}_1$, in d dimensions:

$$\cos \theta_d = \frac{\mathbf{e}_1^T \mathbf{1}}{\|\mathbf{e}_1\| \|\mathbf{1}\|} = \frac{\mathbf{e}_1^T \mathbf{1}}{\sqrt{\mathbf{e}_1^T \mathbf{e}_1} \sqrt{\mathbf{1}^T \mathbf{1}}} = \frac{1}{\sqrt{1} \sqrt{d}} = \frac{1}{\sqrt{d}}$$

Diagonals in Hyperspace

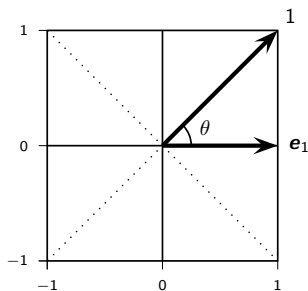
As d increases, we have

$$\lim_{d \rightarrow \infty} \cos \theta_d = \lim_{d \rightarrow \infty} \frac{1}{\sqrt{d}} \rightarrow 0$$

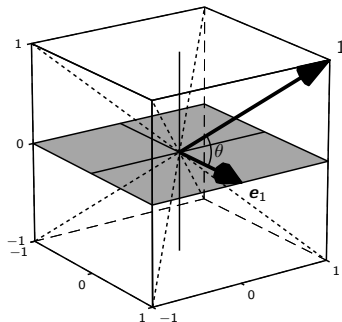
which implies that

$$\lim_{d \rightarrow \infty} \theta_d \rightarrow \frac{\pi}{2} = 90^\circ$$

Angle between Diagonal Vector 1 and e_1



(a) In 2D



(b) In 3D

In high dimensions all of the diagonal vectors are perpendicular (or orthogonal) to all the coordinates axes! Each of the 2^{d-1} new axes connecting pairs of 2^d corners are essentially orthogonal to all of the d principal coordinate axes! Thus, in effect, high-dimensional space has an exponential number of orthogonal “axes.”

Density of the Multivariate Normal

Consider the standard multivariate normal distribution with $\mu = 0$, and $\Sigma = I$

$$f(\mathbf{x}) = \frac{1}{(\sqrt{2\pi})^d} \exp \left\{ -\frac{\mathbf{x}^T \mathbf{x}}{2} \right\}$$

The peak of the density is at the mean. Consider the set of points $\mathbf{x}$ with density at least α fraction of the density at the mean

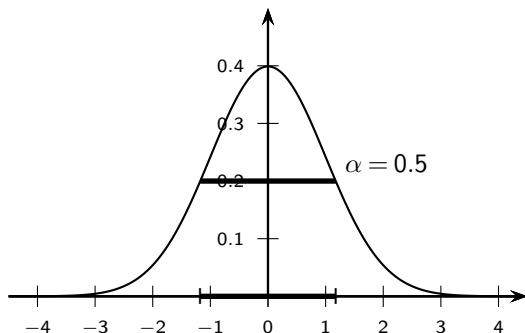
$$\begin{aligned} \frac{f(\mathbf{x})}{f(0)} &\geq \alpha \\ \exp \left\{ -\frac{\mathbf{x}^T \mathbf{x}}{2} \right\} &\geq \alpha \\ \mathbf{x}^T \mathbf{x} &\leq -2\ln(\alpha) \\ \sum_{i=1}^d (x_i)^2 &\leq -2\ln(\alpha) \end{aligned}$$

The sum of squared IID random variables follows a chi-squared distribution χ_d^2 . Thus,

$$P \left(\frac{f(\mathbf{x})}{f(0)} \geq \alpha \right) = F_{\chi_d^2}(-2\ln(\alpha))$$

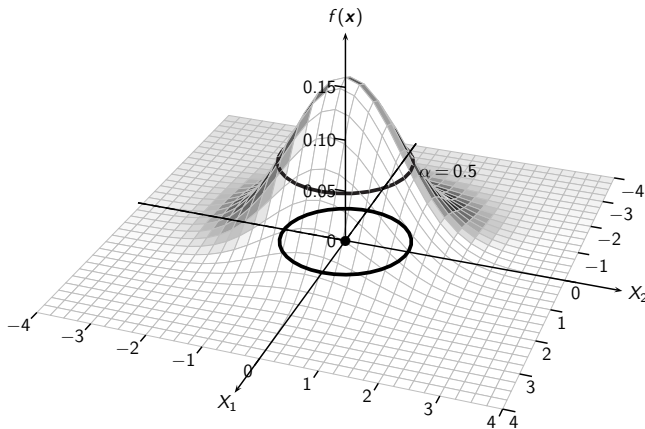
Density Contour for α Fraction of the Density at the Mean: One Dimension

Let $\alpha = 0.5$, then $-2\ln(0.5) = 1.386$ and $F_{\chi^2_1}(1.386) = 0.76$. Thus, 24% of the density is in the tail regions.



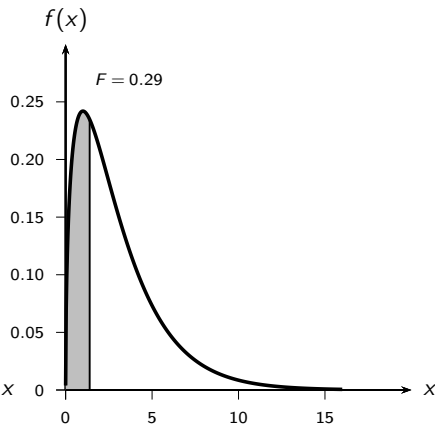
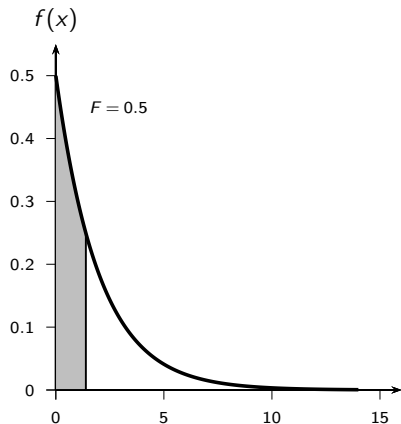
Density Contour for α Fraction of the Density at the Mean: Two Dimensions

Let $\alpha = 0.5$, then $-2\ln(0.5) = 1.386$ and $F_{\chi^2_2}(1.386) = 0.50$. Thus, 50% of the density is in the tail regions.

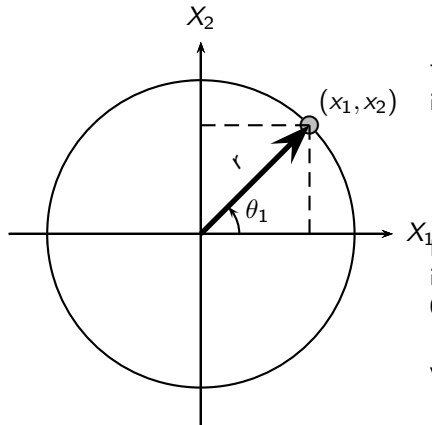


Chi-Squared Distribution: $P(f(\mathbf{x})/f(0) \geq \alpha)$

This probability decreases rapidly with dimensionality. For 2D, it is 0.5. For 3D it is 0.29, ie., 71% of the density is in the tails. By $d = 10$, it decreases to 0.075%, that is, 99.925% of the points lie in the extreme or tail regions.



Hypersphere Volume: Polar Coordinates in 2D



The point $\mathbf{x} = (x_1, x_2)$ in polar coordinates

$$x_1 = r \cos \theta_1 = r c_1$$

$$x_2 = r \sin \theta_1 = r s_1$$

where $r = \|\mathbf{x}\|$, and $\cos \theta_1 = c_1$ and $\sin \theta_1 = s_1$.

The *Jacobian matrix* for this transformation is given as

$$J(\theta_1) = \begin{pmatrix} \frac{\partial x_1}{\partial r} & \frac{\partial x_1}{\partial \theta_1} \\ \frac{\partial x_2}{\partial r} & \frac{\partial x_2}{\partial \theta_1} \end{pmatrix} = \begin{pmatrix} c_1 & -r s_1 \\ s_1 & r c_1 \end{pmatrix}$$

Hypersphere volume is obtained by integration over r and θ_1 (with $r > 0$, and $0 \leq \theta_1 \leq 2\pi$):

$$\begin{aligned} \text{vol}(S_2(r)) &= \int_r \int_{\theta_1} |\det(J(\theta_1))| dr d\theta_1 \\ &= \int_0^r \int_0^{2\pi} r dr d\theta_1 = \int_0^r r dr \int_0^{2\pi} d\theta_1 \\ &= \frac{r^2}{2} \Big|_0^r \cdot \theta_1 \Big|_0^{2\pi} = \pi r^2 \end{aligned}$$

Hypersphere Volume: Polar Coordinates in 3D

$\mathbf{x} = (x_1, x_2, x_3)$ in polar coordinates

$$x_1 = r \cos \theta_1 \cos \theta_2 = r c_1 c_2$$

$$x_2 = r \cos \theta_1 \sin \theta_2 = r c_1 s_2$$

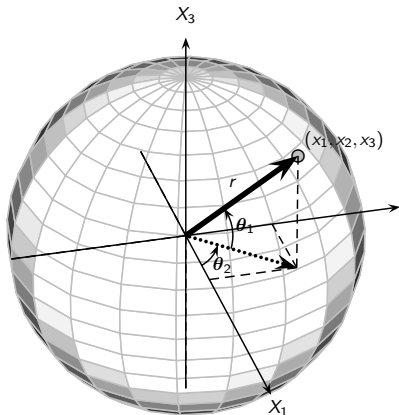
$$x_3 = r \sin \theta_1 = r s_1$$

The Jacobian matrix is given as

$$J(\theta_1, \theta_2) = \begin{pmatrix} c_1 c_2 & -r s_1 c_2 & -r c_1 s_2 \\ c_1 s_2 & -r s_1 s_2 & r c_1 c_2 \\ s_1 & r c_1 & 0 \end{pmatrix}$$

The volume of the hypersphere for $d = 3$ is obtained via a triple integral with $r > 0$, $-\pi/2 \leq \theta_1 \leq \pi/2$, and $0 \leq \theta_2 \leq 2\pi$

$$\begin{aligned} \text{vol}(S_3(r)) &= \int_r \int_{\theta_1} \int_{\theta_2} |\det(J(\theta_1, \theta_2))| dr d\theta_1 d\theta_2 \\ &= \frac{4}{3} \pi r^3 \end{aligned}$$



Hypersphere Volume in d Dimensions

The determinant of the d -dimensional Jacobian matrix is

$$\det(J(\theta_1, \theta_2, \dots, \theta_{d-1})) = (-1)^d r^{d-1} c_1^{d-2} c_2^{d-3} \dots c_{d-2}$$

The volume of the hypersphere is given by the d -dimensional integral with $r > 0$, $-\pi/2 \leq \theta_i \leq \pi/2$ for all $i = 1, \dots, d-2$, and $0 \leq \theta_{d-1} \leq 2\pi$:

$$\begin{aligned} \text{vol}(S_d(r)) &= \int_r \int_{\theta_1} \int_{\theta_2} \dots \int_{\theta_{d-1}} \left| \det(J(\theta_1, \theta_2, \dots, \theta_{d-1})) \right| dr d\theta_1 d\theta_2 \dots d\theta_{d-1} \\ &= \int_0^r r^{d-1} dr \int_{-\pi/2}^{\pi/2} c_1^{d-2} d\theta_1 \dots \int_{-\pi/2}^{\pi/2} c_{d-2} d\theta_{d-2} \int_0^{2\pi} d\theta_{d-1} \\ &= \frac{r^d}{d} \frac{\Gamma\left(\frac{d-1}{2}\right) \Gamma\left(\frac{1}{2}\right)}{\Gamma\left(\frac{d}{2}\right)} \frac{\Gamma\left(\frac{d-2}{2}\right) \Gamma\left(\frac{1}{2}\right)}{\Gamma\left(\frac{d-1}{2}\right)} \dots \frac{\Gamma(1) \Gamma\left(\frac{1}{2}\right)}{\Gamma\left(\frac{3}{2}\right)} 2\pi \\ &= \frac{\pi \Gamma\left(\frac{1}{2}\right)^{d/2-1} r^d}{\frac{d}{2} \Gamma\left(\frac{d}{2}\right)} \\ &= \left(\frac{\pi^{d/2}}{\Gamma\left(\frac{d}{2} + 1\right)} \right) r^d \end{aligned}$$

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